Exploratory Data Analysis (EDA)– Delinquency Prediction

1. Introduction

The purpose of this report is to conduct an Exploratory Data Analysis (EDA) on the Geldium customer financial dataset to identify patterns, data quality issues, and potential risk indicators related to delinquency. The analysis focuses on understanding the dataset structure, detecting missing or inconsistent values, and exploring relationships between financial and behavioral variables that may influence delinquency outcomes.

Through this analysis, key variables such as credit utilization, missed payments, income, and debt-to-income ratio are examined to determine their potential impact on predicting delinquent accounts. The insights generated from this EDA will help ensure the dataset is properly prepared for predictive modeling and support the development of accurate delinquency risk prediction models.

2. Dataset Overview

- **Total records:** 500 customers
- **Total variables:** 19
- **Data types:**
  - Numerical: Age, Income, Credit Score, Credit Utilization, Loan Balance, Debt-to-Income Ratio, Account Tenure
  - Categorical: Employment Status, Credit Card Type, Location
  - Behavioral/payment history: Month_1 – Month_6
  - Target variable: Delinquent_Account (0 = No, 1 = Yes)

3. Missing Data Analysis

Notable Missing or Inconsistent Data

- **Income:** 39 missing values (~7.8%)
- **Loan_Balance:** 29 missing values (~5.8%)
- **Credit_Score:** 2 missing values
- Missing values in financial variables could bias risk predictions if not handled properly.

4. Key Findings and Risk Indicators

Key Anomalies:

- **Credit Utilization values exceed 100%** (max ≈ 102.6%), which may indicate:
    - Data entry errors
    - Customers exceeding credit limits
- **Payment history fields** show frequent "Late" and "Missed" values in several months.
- **Class imbalance:**
    - Non-delinquent: 420
    - Delinquent: 80
      This imbalance could impact predictive modeling.

Early Indicators of Delinquency Risk:

- **Missed_Payments:** Customers with more missed payments likely have higher delinquency risk.
- **Credit_Utilization:** Higher utilization indicates financial stress.
- **Debt_to_Income_Ratio:** Higher ratios suggest greater debt burden.
- **Recent payment behavior (Month_1–Month_6):** Repeated late or missed payments signal repayment problems.

5. AI & GenAI Usage

- GenAI tools were used to **summarize the dataset structure, detect missing values, and identify anomalies**.
- AI assisted in **suggesting imputation strategies** for missing financial variables such as income and loan balance.
- GenAI helped **identify key variables likely to predict delinquency**, including missed payments, credit utilization, and debt-to-income ratio.
- AI-generated insights supported **faster pattern detection and preliminary risk assessment** during exploratory data analysis.

Example prompts used during the analysis include:

- "Summarize key patterns, outliers, and missing values in this dataset. Highlight any fields that might present problems for modeling delinquency."
- "Identify the top variables most likely to predict delinquency based on this dataset and explain why."
- "Suggest an imputation strategy for missing financial variables such as income or loan balance based on industry best practices."
- "Detect patterns in payment history that may indicate delinquency risk."

6. Conclusion & Next Steps

- EDA identified **key delinquency risk indicators**, including missed payments, high credit utilization, high debt-to-income ratio, and low credit scores.
- The dataset contained **missing values in income, loan balance, and credit score**, which were addressed using appropriate imputation methods.
- Some **data anomalies**, such as credit utilization values above 100%, were identified and flagged for further review.
- Insights from this analysis help **prepare the dataset for predictive modeling**.

Next Steps:

- Decide the best way to deal with missing income and loan balance values, ensuring that the chosen method does not introduce bias.
- Double-check whether high credit utilization and missed payments remain the strongest indicators of delinquency across different customer groups.
- Look into records where customers have high income but low credit scores to see if there are reporting errors or other explanations.

.